BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. It references a candidate/primary key in a related table.

Q2. O(log n) for sorted random-access data.

Q3. Read-only memory.

Q4. The longest-waiting item at the front.

### You Do pages · Independent questions

Q1. 1:M definition/example [3], M:N definition/example [3].

Q2. Author(PK) [2], Book(PK) [2], Authorship with two FKs/composite PK [2].

Q3. Atomicity/1NF problem [2], hard joins/search [2], integrity/update anomaly [2], junction table solution and cardinality [2].

### Exit Ticket · Model answer

False. A junction table is normally required.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 135 · Lesson 135 · Entity Relationships: One-to-Many and Many-to-Many

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the role of a foreign key?

Q2. In Big O notation, what is binary search worst-case complexity?

Q3. What does ROM stand for?

Q4. Which is removed first from a FIFO queue?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

entity — An identifiable class of thing stored in database.

relationship — Association between entities.

cardinality — How many related instances are possible.

junction table — Table breaking a many-to-many relation into two one-to-many relations.

### New learning

Identify entities — Find objects with independent identity, not every noun.

Set cardinality — One student takes many courses, and each course has many students.

Resolve M:N — Add Enrolment table referencing StudentID and CourseID.

### Worked example

Represent students enrolling on several courses, with many students per course.

1. Student↔Course is M:N — Find the given data and name the required result.

2. Create Enrolment bridge — Explain why this intermediate stage is necessary.

3. Two one-to-many links — Student(StudentID PK) and Course(CourseID PK) form many-to-many. Resolve with Enrolment(StudentID FK, CourseID FK, EnrolledOn), with composite PK (StudentID, CourseID) if enrolment is unique per pair.

## We Do · Practise with guidance

A customer can place multiple orders; an order belongs to one customer. Model the relationship.

One Customer

Find the given data and name the required result.

Many Orders

Explain why this intermediate stage is necessary.

FK in Order table

Customer 1:M Order. CustomerID PK in Customer; CustomerID FK in Order. OrderID serves as the Order PK. Optionality (must a customer have an order?) needs a business rule.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain the difference between one-to-many and many-to-many relations with examples. [6]

### Question 2

Design tables for Author and Book when each book has multiple authors. [6]

### Question 3

Evaluate why a field containing comma-separated ClubIDs is poorer than a Membership junction table. [8]

## Exit Ticket · Answer before checking

A many-to-many relationship can always be represented correctly with just a single FK on one of the two entity tables.